

- A list of characteristics to be addressed in the validation experiment, such as precision, accuracy, applicability, very often including selectivity, response function and calibration, etc.
- One or more parameters for each of these characteristics are aimed for providing quantitative estimates, such as standard deviation, recovery rate, and percentages.

If no acceptance criteria are explicitly set for evaluating the estimated values of the parameters, the analyst only performs a characterization of the method, which is not a full validation. To be complete, the values of the parameters must be checked as conforming to performance criteria values. In practice, the statistical approach proposed to estimate parameters is simple numeric calculations, such as average, standard deviation, or the coefficient estimators of a function obtained by using the least-squares regression method. For qualitative methods, it is a matter of proportions, with confidence intervals obtained by referring, most often, to a binomial distribution. Therefore, all these calculations can easily be done with simple software, such as Microsoft Excel or Open Office Calc.

A more recent trend is to apply more elaborate algorithms, such as the weighted least-squares (WLS) method or partial least-squares (PLS) regression (Section 2.3). These methods require computation procedures unavailable in worksheets. In many analytical devices, such as mass spectrometers or near infra-red spectrometers, manufacturers have included them directly in the monitoring software. But, the Python language can help the analysts needing to develop their solutions.

The diverse estimators proposed in different validation guides to estimate the same parameter is another confusing issue for analysts, such as robust algorithms to estimate reproducibility [13]. Other computational methods requiring intensive calculation, such as Monte Carlo simulation, remain in the domain of specialized scientific publications in analytical sciences but are usual in other measurement fields. This specific topic is not addressed in this book.

Sometimes statistical tests, known as significance tests, are recommended to assess the conformity of a parameter to a criterion [14]. Unfortunately, only the first type of risk of error, denoted α risk, is usually considered. This is also known as the producer's risk or the risk for an analyst who used a valid method but concludes that it was not. In contrast, the customer's risk, denoted β -risk, i.e. the risk to the client that the method is not valid but regarded as valid by the analyst, is not assessed.

It is exceptional that the validation procedure indicates how to organize the experimental design. This is left to the experimenter's initiative, who may proceed randomly and intuitively. This topic is discussed in more detail in Section 5.4.2. While a list of characteristics and/or parameters to be experimentally estimated is required, the analyst remains free to choose the experimental design that is considered the most appropriate to achieve this goal.

The classic validation procedure can be called a multicriteria method because it consists of evaluating several characteristics separately compared to a set of criteria and making individual decisions on each. Therefore, contradictory conclusions may occur, such as conforming precision but non-conforming trueness. The final decision is difficult to make. While a list of characteristics and/or parameters to be

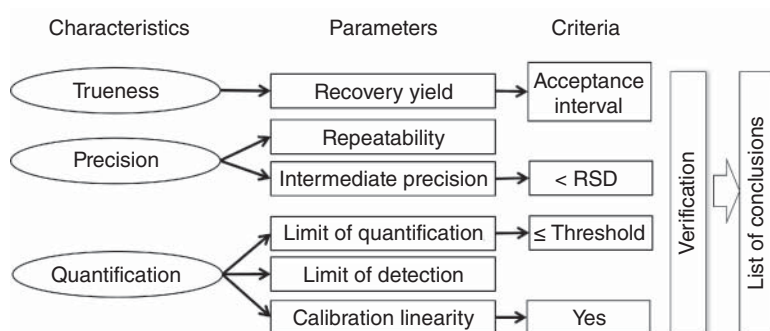


Figure 5.2 Example of a multicriteria validation procedure.

experimentally estimated is required, the analyst is free to choose the experimental design that is considered the most appropriate to achieve this goal.

As illustrated in Figure 5.2, based on the characteristics and parameters required in ISO 17025 standard, the classical validation strategy can be described as a *multicriteria* method because it consists of evaluating the different characteristics sequentially and separately against a set of criteria and making an individual decision on each one [15].

This strategy may provide a lot of interesting information to the analyst, but practice shows that it has the disadvantage of leading to a set of conclusions that may be contradictory. For example, precision is conforming, while trueness is not. How to conclude? The final decision may quickly become subjective. But the major downside is that it is not evident that this knowledge is the proper answer to “satisfying customer needs” as claimed in quality assurance standards.

For instance, US-FDA has set up a list of acceptance criteria to define a validated method called standard method performance requirement or SMPR [16]. This document has a set of criteria values for precision and recovery yield. These values are included in several other official documents issued by different institutions, such as the European Commission or the Codex Alimentarius [1, 10].

But, when comparing several guides applicable at the international level for regulatory or health purposes, it is surprising that in many cases, no acceptance criteria nor verification procedures are proposed. The number of degrees of freedom available for decision-making is generally ignored. However, it is crucial when statistical testing is required, as underlined in many statistical handbooks.

Alternatively to the classic strategy, it is possible to propose a validation procedure based on a unique criterion. The latter amounts to choosing a unique comprehensive characteristic which may be a combination of several parameters. Then by comparing this parameter combination to a single criterion, it is possible to obtain more easily a final decision. Such single-criterion validation procedure are less often used. A possible example is using the total analytical error (*TAE*) described below.

But it is also appropriate to select accuracy as a unique validation characteristic, which is defined as the “closeness of agreement between a measured value and a true

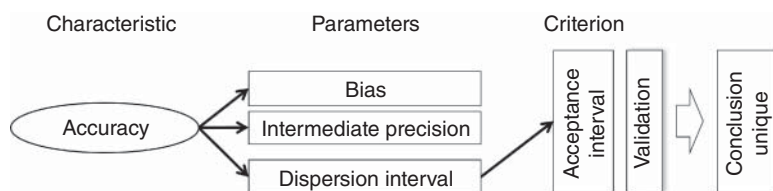


Figure 5.3 Example of a single-criterion validation procedure.

value of a measurand” and aims to verify if this closeness is satisfactory. A detailed discussion about the concept of accuracy in this context is given in Section 6.10.

To mathematically express accuracy, it is promising to combine several parameters into one statistical parameter, for instance, one so-called “statistical dispersion interval.” This lesser-known family of statistical intervals is described in Section 5.3. The latter is compared to an acceptance interval, resulting in a unique parameter *versus* a unique criterion.

The MAP is a validation procedure described in Section 5.2 that is an example of a single-criterion strategy based on accuracy, slightly different from *TAE*. It was developed in the late 1990s, in the framework of a commission of the Société française des sciences et techniques pharmaceutiques (SFSTP). It was published in a series of papers [17–20]. Figure 5.3 illustrates schematically this single-criterion validation procedure.

This method, applied for nearly 20 years in many laboratories, on a wide range of methods, and in varied scientific contexts (Section 5.1), was adopted by several international standards addressing method validation, such as ISO-16140 or ISO 22116.

MAP was chosen to describe this procedure and accuracy as a unique characteristic because it is often identified as the combination of trueness and precision (even the correct “combination” to be used is undefined). For example the ISO 5725 standards have the general title: “Accuracy (trueness and precision) of measurement results and methods.” It seemed logical to estimate accuracy by combining two parameters, the first for trueness, such as a bias or a recovery rate, and the second for precision, such as a standard deviation of precision.

A comparable combination was already proposed in the 1970s to introduce the concept of total analytical error, or *TAE*, thoroughly described in the first Section 6.1. of Chapter 6 [21]. The recently revised definition of accuracy [22] explains that it is a “non-numerical” concept. The initial choice seems inconsistent. Nevertheless, for convenience and because it is now well-known, the term method accuracy profile is retained here. There is also a possible confusion with the use of accuracy in some documents: accuracy is not trueness as explained in Chapter 4, although both definitions seem relatively close, as visible in the glossary at the end of the book (see Glossary of Used Terms).

The estimation of the precision is achieved by means of a standard deviation. As explained in Section 2.4.3, ISO 5725 standards define at least three experimental conditions for computing a precision standard deviation, namely reproducibility, repeatability, and in-between intermediate precision. The reproducibility

condition is considered well-adapted to interlaboratory validation studies. But the intermediate precision condition is most adequate for in-house validation because it allows accounting for various sources of laboratory-specific variation.

As described in Section 7.2, intermediate precision is best adapted to a simple and direct estimation of a large part of measurement uncertainty (MU). It is recognized that this goal is out of reach under repeatability conditions. With hindsight and experience, when comparing classic procedure and MAP, some major advantages can be stressed, as MAP permits:

- To compute statistical dispersion intervals (Section 5.3) that give an insight into the measurement value scattering and to verify that it complies with the method acceptance performance.
- To yield a comprehensive procedure in estimating MU as justified in Section 7.2.
- To apply a consistent experimental design which helps verify if the number of trials is optimized as discussed in Section 8.4.3.
- Finally generate the method uncertainty function applicable to any future sample analyzed at the laboratory as explained in Section 7.5.

For these reasons, the MAP is used as a roadmap throughout the whole book and presented as an adequate solution to many problems the laboratory faces for validating a method and finally to have access to a relatively simple estimation procedure of MU.

5.2 Method Accuracy Profile (MAP)

5.2.1 Principles

Because incorrect measurements can lead to wrong decisions and, therefore, to significant additional costs or even risks, MAP is a validation procedure founded on the following reasoning: to validate a method means to demonstrate that it can produce an important, even prerequisite, proportion of measurement values serviceable for a correct decision, for example at least 80% or 90% of possible measures are distributed within an acceptance interval around the true value of a sample. Whereas accuracy is defined as the “closeness of agreement between a measured value and a true value of a measurand” this parameter can generate confidence in the analytical result, once estimated.

The acceptance interval is specified as $[A_L; A_U]$, preferably pre-defined by the analysis end-user. The subscript L of A_L stands for lower bound, and U for upper bound. In other words, the analyst can claim the method as validated, with a known confidence level, when it is able to produce at least the required proportion of data lying within the acceptance interval bounds. The major innovation of the suggested method is to use statistical dispersion intervals to compute how much acceptable measures are scattered. These lesser known statistics are estimators of the probable dispersion of results and are expressed as intervals.

Ultimately, this approach is more straightforward than computing separate statistical parameters, such as means or variances, and independently checking

whether they conform. Consequently, in statistical terms, to demonstrate validity and check whether the method fulfills defined requirements, the proposed approach is to calculate the statistical dispersion interval within which lies the required proportion of measurements and verify it is included within the acceptance interval. As explained below, several types of statistical dispersion intervals exist and can be applied to this goal. Starting from this reasoning, the MAP harmonized procedure was established and published between 2004 and 2008 in the *Journal of Pharmaceutical and Biomedical Analysis* [17–20].

A mnemonic presentation of MAP consists of the 10-step list registered in The 10-step MAP procedure provided at the end of the book. The importance of the concept of *series* indicated at step 4, was already underlined in Section 3.1 about inter-laboratory studies. A series includes all measurements performed under identical conditions, for example, the same method, the same day, the same operator, the same calibration curve, etc. This reflects the usual laboratory management where samples submitted to the same analysis are grouped before getting started. This organization aims to reduce the fixed costs of an analytical operating procedure as explained in Section 4.4 about control charts. Because a method must be validated the way it will be used in the future, the practical layout of the series should reflect as closely as possible the sources of uncertainty likely to occur during routine application of the method. If the measurements are not organized into series, it is impossible to complete the calculations and construct the MAP. The requisite of the experimental design is explained in Section 5.4.2.

In the 1990s, some guidelines were proposed to validate a method for a single concentration level. But it was quickly realized that the figures of merit of a method – precision and trueness – vary with the concentration. Therefore, it is now considered mandatory to perform validation measurements on at least three validation materials of different concentrations spreading over the validation range.

Once measurements are collected and inverse-predicted concentrations are obtained for each material, the bounds of β -expectation tolerance intervals (β -ETI) are calculated by combining the different validation parameters. Statistical dispersion intervals consist of a family of statistical parameters less well-known than confidence intervals, thus extensively explained in Section 5.3.1. They are designed to contain a certain proportion, denoted β , of future outcomes or sometimes a single future outcome.

The letter β is the coverage probability of the interval and corresponds to this proportion. In the text, it is denoted β when expressed between 0 and 1, such as 0.80, or $\beta\%$ as a percentage, such as 80%. These two equivalent notations are a reminder that in worksheets, it must be distinguished between the number, as stored in computer memory, and its display format, as visible in a worksheet cell.

Finally, for a given validation material, the β -ETI corresponds to an interval where it can be predicted that a proportion $\beta\%$ of future measurement values may fall. By convention, the coverage probability is denoted β and should not be confused with the risk of error β of hypothesis testing. It can be expressed as:

Generic expression of the statistical dispersion interval

$$[\bar{Z} \pm k_{TI} \times s_{TI}] \quad (5.1)$$